

BONAM MINGOLE

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Ph.D. Student ◇ AI for Social Good ◇ University Park, PA, USA

EDUCATION

Ph.D. in Informatics Pennsylvania State University GPA: 3.96 <i>Research focus:</i> Uncovering Socio-Technical Limitations in LLMs <i>Advisor:</i> Prof. Amulya Yadav	State College, USA <i>Aug. 2024 - Present</i>
M.S. in Software Engineering University of Quebec in Montreal GPA: 3.67 <i>MS Thesis:</i> Analysis and co-creation of an AI assistant in the Financial Industry	Montreal, Canada <i>Sept. 2017 - Jan. 2020</i>
B.B.A in Business Technologies University of Quebec in Montreal GPA: 3.41 <i>International exchange student:</i> EMS Strasbourg, France.	Montreal, Canada <i>Sept. 2013 - May 2017</i>

PUBLICATIONS

6. **Mingole, B.**, Majumdar, A., Choudhury, F. A., Kraschnewski, J. L., Sundar, S. S., & Yadav, A. (2025). Dr. GPT Will See You Now, but Should It? Exploring the Benefits and Harms of Large Language Models in Medical Diagnosis using Crowdsourced Clinical Cases. In Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency. [FAccT 2026](#).
5. Guo, H., Venkit, P.N., Jang, E., Srinath, M., Zhang, W., **Mingole, B.**, Gupta, V., Varshney, K.R., Sundar, S.S. and Yadav, A. (2025). *Exposing AI Bias by Crowdsourcing: Democratizing Critique of Large Language Models*. In Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (Vol. 8, No. 2, pp. 1169-1180). [AIES 2025](#).
4. Wang, X., Zhang, W., Koneru, S., Guo, H., **Mingole, B.**, Sundar, S. S., Rajtmajer, S. & Yadav, A. (2025). *Have LLMs reopened the Pandora's Box of AI-generated fake news?*. In Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers) (pp. 2795-2811). [NAACL 2025](#).
3. Rismani, S., Davis, L., **Mingole, B.**, Rostamzadeh, N., Shelby, R., & Moon, A. (2025). *Responsible AI measures dataset for ethics evaluation of AI systems*. [Nature Scientific Data 2025](#).
2. **Mingole, B.**, Gilmour, A., Diallo D., Patel, J., Mussio, M. (2023). *AITHICAL: an AI Ethics guidelines aggregator Prototype*. [AI4Good Lab 2023](#).
1. **Mingole, B.** (2020). Analysis and Co-creation of an AI assistant in the financial industry [Master's thesis, University of Quebec in Montreal]. [Archipel UQAM](#).

MEDIA COVERAGE OF RESEARCH

9. Huckins, G. (2026, Jan 22). "Dr. Google" had its issues. Can ChatGPT Health do better? [MIT Technology Review](#).
8. Tutella F. (2025, Nov 4). *Lay intuition as effective at jailbreaking AI chatbots as technical methods, research suggests*. [Tech Xplore](#).
7. Sourwine, A. (2025, Apr 28). *Penn State's 'Cheat-a-Thon' Pinpoints Strengths, Weaknesses of AI*. [GovTech](#).
6. Thiruselvam, M. (2025, Mar 4). *Penn State Set To Host 'Cheat-A-Thon' Showcasing AI's Integration Into Academia*. [Onward State](#).
5. EdScoop Staff (2025, Mar 3). *Penn State begins 'Cheat-a-thon' to test limits of generative AI*. [EdScoop](#).
4. Ford, J. (2024, Nov 4). *Contest explores AI's strengths, flaws for medical diagnoses*. [Penn State University](#).
3. Ford, J. (2024, May 8). *Competition highlights believable fake news created with generative AI tools*. [Penn State University](#).
2. Essien, E. (2024, Feb 1). *Penn State's Bias-a-thon: A Beacon for Ethical AI in the 21st Century*. [The Inclusive AI](#).
1. Pennsylvania State University (2023, Oct 27). *Penn Staters Challenged to Detect Bias in AI Tools*. [Mirage News](#).

RESEARCH EXPERIENCE

AI Research Assistant | RAISE Lab @ Penn State University
Uncovering & Mitigating Socio-Technical Limitations in LLMs

Stage College, USA
Aug. 2024 - Now

- Designed and executed mixed methods research studies investigating LLMs limitations such as bias, misinformation, and their socio-technical impacts in healthcare and education, including mitigation strategies for safer deployment.
- Developed AI-driven adaptive educational AI systems leveraging LLM fine-tuning, retrieval-augmented generation (RAG), and personalized tutoring methodologies. See [ALAASKA Repository](#)
- Built RAG-based medical diagnostic pipelines improving LLM accuracy from 76% to 90%. See [DAT RAG Repository](#).
- Evaluated algorithmic fairness in LLMs through cross-regional performance benchmarking and bias analysis.

Humanitarian AI Research Intern | United Nations OCHA
AI for Humanitarian Analysis and Crisis Response

The Hague, Netherlands
Jun - Aug 2025

- Conducted research on the use of AI tools to support humanitarian analysis for crisis response planning.
- Prototyped finetuned LLMs leveraging RAG and prompt engineering for accurate organizational information retrieval.
- Supported the OCHA-AI community by documenting best practices and AI toolkits for field staff.
- Drafted internal reports and recommendations on responsible AI adoption for humanitarian action.
- Collaborated across UN branches to align AI initiatives with humanitarian principles and data protection standards.

AI Research Assistant | Mila Quebec AI Institute
Responsible AI operationalization through Metrics

Montreal, Canada
Aug. 2023 - Aug. 2024

- Conducted a *Scoping Review* on how AI researchers operationalize ethical principles through measurement.
- Extracted data from 257 papers (from 2011 - 2023) measuring AI systems adherence to Responsible AI principles.
- Created the Responsible AI Measures Dataset, consolidating 11,865 data points across 791 evaluation measures, covering 11 ethical principles including Fairness, Transparency, Privacy, Trust, Sustainability, and more.
- Developed a taxonomy of metrics and evaluation methods for Responsible AI principles.

AI Research Fellow | AI4Good Lab
Applied NLP Research and AI Policy

Montreal, Canada
May 2023 - Aug. 2023

- Developed AITHICAL, a prototype ML platform that aggregates AI ethics guidelines and regulations and suggests regulatory guidance to AI practitioners based on their projects descriptions. See [AITHICAL GitHub Repo](#)
- Trained and tested a zero-shot NLP model to classify AI projects into 6 categories targeted by Canadian AI regulations.
- Initiated a mentorship program to create a more inclusive and diverse Tech culture in Canada.

AI Research Partnerships Coordinator [Part-time] | Mitacs Canada
AI Research Governance

Montreal, Canada
Jan. 2022 - Aug. 2024

- Evaluated 10,000+ AI research grant proposals, contributing to the allocation of \$50M+ in research funding.
- Designed a structured evaluation framework that reduced review time by 20% while maintaining quality.
- Facilitated partnerships between academia, industry, and nonprofits, enabling 130+ human-centered AI initiatives.
- Analyzed proposals trends to identify emerging AI research areas, guiding future funding strategies.

INDUSTRY EXPERIENCE

AI Solutions Analyst | Intact Insurance - Data Lab
Applied Machine Learning and AI deployment

Montreal, Canada
May 2021 - Apr. 2023

- Coordinated the development and deployment of XGBoost regression models to predict insurance premiums for 500K+ commercial and personal policies, achieving 98% model accuracy.
- Deployed NLP solutions to summarize insurance claims, enhancing efficiency for 5k+ employees.
- Conducted A/B testing, model validation, and user acceptance testing ensuring 95% user satisfaction rate.

AI Solutions Analyst | National Bank of Canada
Participatory design for NLP Solutions

Montreal, Canada
Jun. 2018 - Apr. 2021

- Used participatory design to implement an Azure-based chatbot for HR inquiries, serving 7K+ employees across Canada.
- Gathered and managed requirements for a smart search engine, enhancing information retrieval accuracy by 15%.
- Translated AI research findings into actionable insights aligned with organizational digitalization goals.

TECHNICAL SKILLS

Programming Languages	Python, SQL, JavaScript, HTML, Bash
Framework & Tools	LLMs Finetuning, RAG, LangChain, Sci-Kit Learn, Pytorch, React, FastAPI, Git

MACHINE LEARNING (ML) AND AI PROJECTS

DigitalAIze <i>Collection of ML applications to facilitate Digital Transformation</i>	DigitalAIze Web App
Set of Online Machine learning applications aimed at enabling Digital Transformation for small businesses in communities with limited access to information technology.	<i>Oct 2023</i>
MRAW <i>Prediction of Life Expectancy based on Access to Clean Drinking Water</i>	MRAW Web App
Online ML solution using XGboost regression to predict life expectancy in a country based on the percentage of people accessing clean drinking water.	<i>Jul 2022</i>

TEACHING EXPERIENCE

Teaching Assistant	
• DS 442 (PSU) Artificial Intelligence Foundations	<i>Aug. 2025 - Dec. 2025</i>
• INF 2050 (UQAM) Software Development Tools and Practices	<i>Jan. 2019 - May 2019</i>
• INF 1120 (UQAM) Introduction to Programming	<i>Sep. 2018 - Dec. 2018</i>